

Software Engineering & Project Management Documentation

Document 07: Business Requirements Document (BRD)

Document Control & Metadata

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- Author / Principal Business Analyst:** Ritesh Jadhav, Lead Systems Analyst & Solution Architect
- Project Sponsor:** Managing Director & Executive Board, PharmaCart Retail Pharmacy Chain
- Target Audience:** Executive Steering Committee, Store General Managers, Commercial & Clinical Stakeholders, Architecture & Engineering Teams

Table of Contents

- [1. Executive Summary & Business Context](#)
- [2. Problem Statement & Business Opportunity](#)
- [3. Current State \(As-Is\) vs. Future State \(To-Be\) Analysis](#)
- [4. Business Goals, Objectives & Strategic KPIs](#)
- [5. Stakeholder Analysis & Detailed Personas](#)
- [6. High-Level Business Requirements \(BRs\)](#)
- [7. Financial Feasibility, ROI & Business Case](#)
- [8. Business Constraints, Assumptions & Regulatory Boundaries](#)
- [9. Operational Acceptance & Sign-Off](#)

1. Executive Summary & Business Context

1.1 The Retail Pharmacy Landscape in Ahmedabad

PharmaCart operates a successful, established network of **14 physical retail pharmacies** strategically distributed across the metropolitan region of Ahmedabad, Gujarat (covering high-density urban corridors including Navrangpura, Satellite, Vastrapur, Maninagar, Paldi, Bopal, Bodakdev, and Naranpura).

For over two decades, the pharmacy chain has built enduring customer loyalty based on neighbourhood trust, personal relationships between community pharmacists and families, and immediate availability of critical acute and chronic therapeutic medications.

However, rapid digital transformation in urban India has disrupted traditional pharmacy retail:

- Mega-e-pharmacy aggregators (e.g., Tata 1mg, Apollo 24/7, PharmEasy, Netmeds) have aggressively penetrated the Ahmedabad market with mobile app ordering and aggressive discounts.
- The Neighbourhood Pharmacy Advantage:** While central-warehouse aggregators promise home delivery, their typical delivery window is **24 to 48 hours** from distant regional distribution centers. In contrast, PharmaCart possesses **14 local fulfillment nodes** situated within 3 to 5 kilometers of 85% of Ahmedabad's residential population.

- By digitally uniting these 14 physical storefronts into a synchronized, real-time omni-channel platform, PharmaCart can provide **under-2-hour doorstep delivery** and **90-second express in-store counter pickup**, combining digital convenience with physical neighbourhood immediacy.

Mega-Aggregators (1mg/Apollo)

- Distant Central Warehouse
- 24 to 48 Hour Shipping
- Detached Customer Service

VS

PharmaCart Omni-Channel

Model

- 14 Local Ahmedabad Stores
- 2-Hour Express Delivery
- 90-Second Store Counter Pickup
- Trusted Neighbourhood Pharmacist

1.2 Purpose of this Document

This **Business Requirements Document (BRD)** establishes the definitive commercial vision, strategic objectives, business requirements, and economic justification for the PharmaCart platform. It bridges commercial strategy and technical execution, serving as the direct baseline for the technical 01_SRS_Document.md and architectural specifications in 02_UML_Package.md.

2. Problem Statement & Business Opportunity

2.1 The Core Business Problem

Despite generating healthy counter footfall, PharmaCart is experiencing an accelerating loss of market share, customer attrition, and operational friction due to four systemic operational bottlenecks:

Business Bottleneck	Commercial Impact	Quantitative Evidence / Metric
1. Digital Footfall Leakage	Young, working-age urban consumers increasingly avoid physical queues and order recurring chronic medications via mobile apps.	Estimated 18% year-on-year drop in chronic refill footfall across premier Ahmedabad branches (Satellite & Navrangpura).
2. Inventory Blind Spots & Phantom Stockouts	The 14 stores operate isolated on-premise desktop billing machines. Store A has no visibility into Store B's stock, causing lost sales when a customer is turned away despite stock existing 2 km away.	38% of walk-in stockouts could have been fulfilled by an adjacent sister store within a 15-minute radius.
3. Peak-Hour Counter Congestion	Validating handwritten doctor prescriptions at the retail counter during peak evening hours (6:00 PM – 9:00 PM) creates severe customer wait times and counter congestion.	Average counter wait time during peak hours exceeds 14.5 minutes per prescription order .
4. Zero Cross-Store Retention	Customer purchases are tracked only as local receipts on isolated POS machines, preventing centralized customer loyalty, auto-refill alerts, or unified service across Ahmedabad.	Customer lifetime value (LTV) is capped; zero visibility into chain-wide purchase frequency.

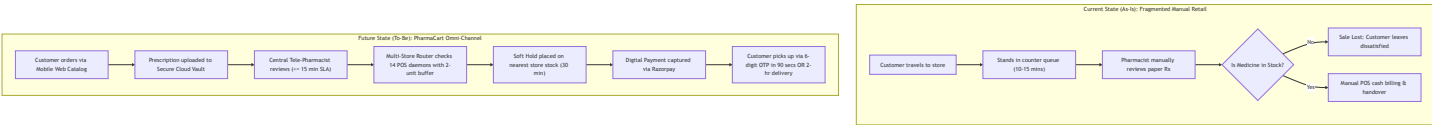
2.2 The Business Opportunity

By developing an integrated, omni-channel commerce and clinical verification platform, PharmaCart can capture a commanding market position:

- 1. **Unlocking Express Hyperlocal Fulfillment:** Leveraging the 14-store footprint to offer guaranteed 2-hour doorstep delivery or 90-second express store counter pickup (Click-and-Collect).
- 2. **Eliminating Stockout Disputes via Safety Stock Logic:** Bridging physical and digital retail by introducing a **2-unit safety stock buffer** ($\text{Effective Available} = \text{Max}(0, \text{Physical Count} - 2)$) and **30-minute temporary soft holds** during customer checkout.
- 3. **Centralizing Tele-Pharmacy Verification:** Moving prescription review from congested physical store counters to a centralized, cloud-based tele-pharmacy desk, achieving a **<= 15 minute clinical review SLA**.
- 4. **Capturing High-LTV Chronic Care:** Establishing customer subscription auto-refills and targeted retail loyalty across diabetes, hypertension, and cardiac patient cohorts.

3. Current State (As-Is) vs. Future State (To-Be) Analysis

3.1 Comparative Process Flow

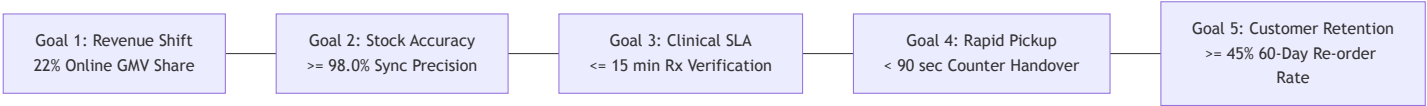


3.2 Operational Capability Delta Matrix

Operational Capability	Current State (As-Is)	Future State (To-Be - PharmaCart Platform)	Strategic Advantage
Customer Ordering Channels	Physical counter walk-in only; occasional informal WhatsApp phone orders.	Progressive Web App (PWA), Mobile Web Catalog, and In-Store Express Kiosk.	24/7 digital order intake across Ahmedabad.
Prescription Review	Dispersed, manual counter review by local store clerk under rush conditions.	Centralized Tele-Pharmacy Review Desk with dual-pane image viewer and audit logging.	Full compliance with GSPC regulations; error rate reduced by >90%.
Inventory Visibility	Disconnected local Windows POS databases (MS SQL, flat files); updated at day-end.	Real-time POS daemon syncing local deltas to cloud aggregator every <= 30 seconds.	Single unified chain inventory pool visible to all customers.
Stockout Prevention	None; physical counter customers frequently take the last box before online fulfillment.	Algorithmic 2-Unit Safety Stock Buffer + 30-Minute Soft Hold on checkout initiation.	Eliminates walk-in vs. online inventory collisions entirely.
Order Routing	Manual guesswork; phone calls between store managers to locate missing drugs.	Automated Proximity Routing Engine allocating orders to the nearest store with confirmed stock.	Minimizes delivery transit time and optimizes store picking workloads.
Fulfillment Options	Counter purchase only.	1. Express Click-and-Collect (OTP Verification). 2. Hyperlocal 2-Hour Delivery.	Omni-channel flexibility tailored to patient urgency.
Customer Data & Retention	Isolated paper slips; zero unified transaction history.	Centralized cloud customer profile with prescription history and repeat re-order triggers.	Foundation for Phase 2 chronic medicine subscriptions (CR-01).

4. Business Goals, Objectives & Strategic KPIs

The PharmaCart leadership committee has established five quantitative business goals to evaluate project success over the initial 6 months post-rollout:



4.1 Key Performance Indicators (KPIs)

Business Goal	Measurable KPI	Baseline (Current)	3-Month Target (Post-Launch)	6-Month Steady State
1. Digital Channel Adoption	Online GMV as % of Total Chain Turnover	0.0%	12.0%	22.0%
2. Inventory Synchronization	Chain-wide stock accuracy rate (Physical vs. Digital)	~72.0% (Estimated)	>= 95.0%	>= 98.0%
3. Clinical Verification Turnaround	Tele-pharmacy prescription review SLA	25 to 45 Mins	<= 20 Mins	<= 15 Mins
4. Click-and-Collect Express SLA	Store counter pickup handover time (OTP verification)	8 to 15 Mins	< 3 Mins	< 90 Seconds
5. Repeat Customer Retention	Chronic prescription 60-day repeat purchase rate	~18.0%	30.0%	>= 45.0%
6. Inventory Discrepancy Rate	Orders rejected due to counter stockout collision	> 14.0%	< 3.0%	< 1.0%

5. Stakeholder Analysis & Detailed Personas

5.1 Primary Stakeholder Matrix

Stakeholder Group	Primary Business Interests	Critical Concerns / Resistance Factors	Success Measure
Executive Leadership (Managing Director & Owner)	Revenue growth, brand defense against 1mg/Apollo, ROI realization, cash runway conservation.	Exceeding the ₹14.0L budget cap; project timeline slipping beyond 5 months.	Delivery of Core MVP within ₹8.80L / 12 Weeks; break-even within 4 months.
Chief Pharmacist & Clinical Team	Legal compliance with Drugs and Cosmetics Act (Schedule H/H1), patient safety, clear audit trails.	Non-pharmacist clerks issuing medicines; regulatory penalties or license suspension.	100% human pharmacist review gate; immutable audit logging for 3 years.
Retail Store Managers (14 Ahmedabad Outlets)	Meeting store sales targets, ease of counter operations, zero counter chaos during evening rush.	Additional data entry burdens; software crashes freezing local counter billing.	Lightweight picking UI; local POS operations completely unblocked by network outages.
Online Retail Customers	Fast access to genuine medicines, quick prescription approval, express store pickup.	Clunky prescription upload; arriving at store only to find the medicine out of stock.	Intuitive mobile web checkout; guaranteed stock hold upon order confirmation.

5.2 Key User Personas

Persona 1: The Chronic Patient Caregiver

- **Name:** Ramesh Bhai Patel (Age 58, Vastrapur, Ahmedabad)
- **Demographics:** Retired civil engineer caring for elderly mother with Type-2 diabetes and hypertension.
- **Pain Point:** Visits the Vastrapur outlet twice a month. Often waits 15 minutes in line only to discover that the specific brand of insulin cartridge is out of stock, forcing him to drive to Navrangpura.
- **Needs from PharmaCart:** A simple mobile web interface to upload his mother's prescription, view guaranteed real-time stock availability, reserve medicines with a 30-minute hold, and collect the pre-packed bag from the counter in under 2 minutes.

Persona 2: The Physical Store Pharmacist

- **Name:** Sandeep Joshi (Age 32, Registered Pharmacist, Satellite Branch)
- **Demographics:** B.Pharm graduate managing physical billing and customer consultation at a high-volume outlet.
- **Pain Point:** During peak evening rush, walk-in customers demand fast billing while online order dispatches require phone coordination. He fears stock being taken by online buyers before counter buyers pay.
- **Needs from PharmaCart:** A background POS daemon that automatically preserves a **2-unit safety stock buffer** for his walk-in customers, coupled with a simple, touch-screen pickup counter terminal that releases pre-bagged orders upon typing a 6-digit OTP.

Persona 3: The Central Tele-Pharmacist Reviewer

- **Name:** Dr. A. K. Patel (VP Clinical & Chief Pharmacist)
- **Demographics:** Senior licensed pharmacist overseeing compliance across all 14 stores.
- **Pain Point:** Reviewing illegible prescription photos on smartphones is error-prone and legally precarious under Gujarat State Pharmacy Council mandates.
- **Needs from PharmaCart:** A dual-pane clinical verification portal displaying high-resolution prescription images on the left and extracted medicine search/dosage pickers on the right, enforcing digital signing and license logging before orders are routed.

6. High-Level Business Requirements (BRs)

The PharmaCart platform must deliver the following core business capabilities:

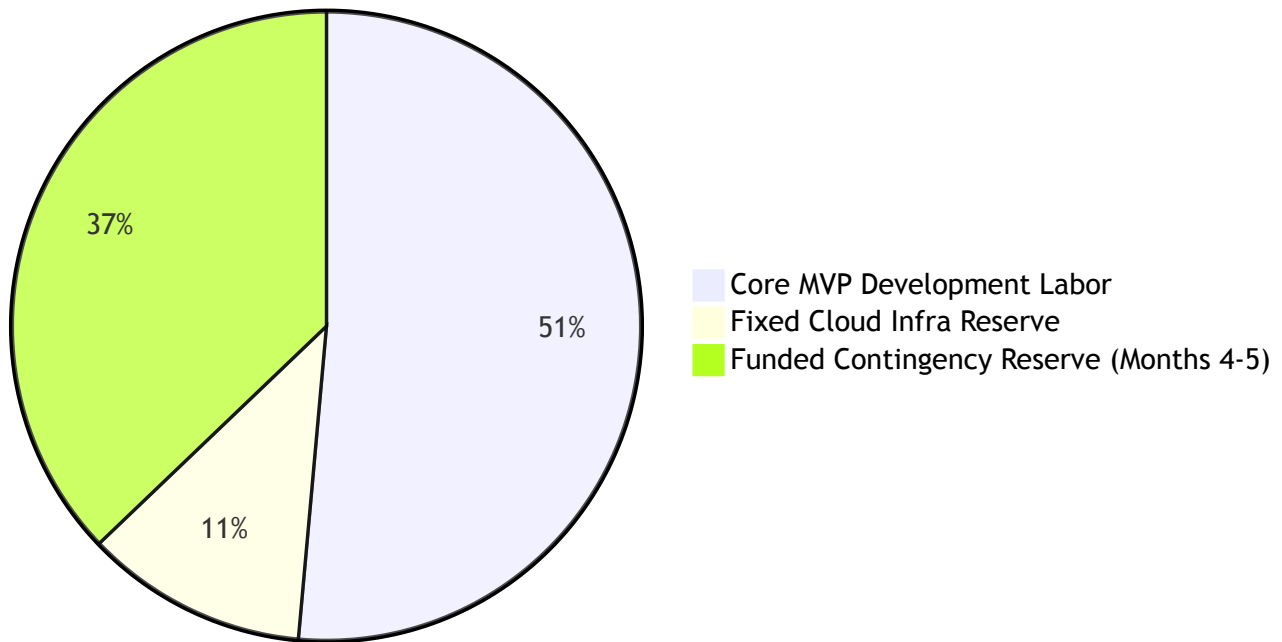
BR Code	Requirement Title	Business Description & Operational Value	Priority (MoSCoW)
BR-01	Digital Medicine Catalog & Search	Customers must be able to search all OTC and prescription medicines, view accurate generic compositions, pricing, and pack sizes without mandatory registration.	Must Have
BR-02	Compliant Prescription Digital Upload	Customers must be able to upload valid prescription images (PDF, JPG, PNG up to 10MB) stored in an AES-256 encrypted vault, triggering tele-pharmacy verification.	Must Have
BR-03	Central Tele-Pharmacy Clinical Review	A centralized queue for licensed pharmacists to inspect prescriptions, approve/reject items, annotate partial fulfills, and digitally sign orders within a 15-minute SLA.	Must Have
BR-04	14-Store Real-Time POS Sync	A lightweight local sync daemon operating on store Windows POS systems to transmit stock additions and counter sales to the cloud within 30 seconds.	Must Have
BR-05	Anti-Mismatch Stock Buffer Engine	Platform must deduct a 2-unit safety buffer from physical stock before displaying online availability, and place a 30-minute soft reservation hold on checkout.	Must Have
BR-06	Proximity-Based Order Routing	Automatically assign orders to the nearest store possessing verified available inventory, with sister-store failover if primary stock is exhausted.	Must Have
BR-07	In-Store Express Pickup & OTP Flow	Store counter pickup workflow enabling customers to retrieve packed orders in under 90 seconds by verifying a cryptographic 6-digit SMS OTP.	Must Have
BR-08	Digital Payment Gateway Integration	Support UPI (Google Pay, PhonePe, Paytm), Credit/Debit Cards, NetBanking via Razorpay with automated webhook reconciliation and instant failure refunds.	Must Have
BR-09	Regulatory Compliance & Audit Logging	Immutable, append-only audit trail logging pharmacist registration numbers, timestamps, prescription hashes, and order states, archived for 3 years.	Must Have
BR-10	Post-Launch Auto-Refill Subscriptions	Recurring scheduled refills for chronic care patients with mandatory pharmacist re-authorization before every monthly dispatch (Phase 2 / CR-01).	Should Have

7. Financial Feasibility, ROI & Business Case

7.1 Capital Allocation & Investment Model

The PharmaCart initiative operates under a hard capital ceiling of **₹14.00 Lakh** allocated by the chain's executive board.

Total Capital Allocation (₹14.00 Lakh Total)



- **Core MVP Execution Cost (Month 1 to 3 / Sprints 1 to 6):**
 - Software Development Labor: 3 Months * ₹2,40,000 / month = ₹7,20,000
 - Infrastructure, Security, AWS Hosting & Payment Gateway Setup: ₹1,60,000
 - **Total Core MVP Investment (Release 1.0): ₹8,80,000**
- **Preserved Cash Contingency Reserve:**

$$\text{Contingency Reserve} = \text{Total Capital (₹14,00,000)} - \text{MVP Cost (₹8,80,000)} = ₹5,20,000$$

This ₹5.20 Lakh reserve provides **2.16 months of fully funded operating runway** (Months 4 and 5) to absorb pilot hardware upgrades, pharmacist training, and the Phase 2 Chronic Care Subscription engine (CR-01).

7.2 Projected Revenue & Break-Even Modeling

Unit Economics & Order Volume Projections across 14 Stores:

- **Average Daily Online Orders per Store (Month 3 to 6):** 15 orders/day/store
- **Total Daily Online Orders across 14 Stores:** 14 * 15 = 210 orders/day
- **Average Order Value (AOV):** ₹650 (Weighted mix of acute and chronic basket sizes)
- **Daily Gross Merchandise Value (GMV):** 210 * ₹650 = ₹1,36,500 / day
- **Monthly Online Gross Turnover (30 Days):** ₹40,95,000 / month (~₹41.0 Lakh)
- **Average Pharmacy Retail Gross Margin:** 18.0%
- **Monthly Gross Profit Generated by Online Channel:**

$$\text{Monthly Gross Profit} = ₹40,95,000 * 18.0\% = ₹7,37,100 / \text{month}$$

- **Net Contribution Margin:** Deducting cloud hosting (₹25,000/mo), payment gateway fees (1.8% = ₹73,710/mo), and local delivery packaging (₹30/order = ₹1,89,000/mo):

$$\text{Net Operating Monthly Profit} = ₹7,37,100 - (₹25,000 + ₹73,710 + ₹1,89,000) = ₹4,49,390 / \text{month}$$

Break-Even Payback Period:

$$\text{Payback Period} = \text{Core MVP Investment (₹8,80,000)} / \text{Net Monthly Profit (₹4,49,390)} = 1.95 \text{ Months of Full Operation}$$

Even under conservative sensitivity modeling where initial online adoption reaches only 50% of targets (105 orders/day), net monthly profit exceeds ₹2,24,000/month, achieving full capital payback within 3.9 months of chain-wide rollout.

8. Business Constraints, Assumptions & Regulatory Boundaries

8.1 Regulatory & Legal Boundaries (Republic of India)

- **The Drugs and Cosmetics Act, 1940 & Rules, 1945:**
 - Schedule H and Schedule H1 drugs (e.g., antibiotics, psychotropics) **strictly require a valid doctor's prescription.** Online dispensing without human pharmacist review is a non-bailable offense.
 - Prescriptions must bear the registered medical practitioner's council number, patient name, date, and doctor's signature.
- **Gujarat State Pharmacy Council (GSPC) Directives:**
 - A registered pharmacist with an active council license must supervise dispensing at every physical node. Centralized review does not waive the physical store pharmacist's final verification responsibility.
- **Digital Personal Data Protection (DPDP) Act, 2023 & IT Act, 2000:**
 - Patient medical records and prescriptions constitute Sensitive Personal Data (SPD) and must be stored within Indian geographic territory under AES-256 encryption.

8.2 Operational & Technical Assumptions

1. **Store POS Hardware Heterogeneity:** The 14 stores operate diverse billing software (primarily Windows-based desktop terminals). The sync agent must be a non-invasive, lightweight background service requiring zero modifications to existing billing databases.
2. **Broadband Network Availability:** While all 14 stores have broadband, occasional localized disconnections occur. The store POS must function autonomously offline without halting physical counter billing.
3. **Delivery Radius:** Express 2-hour doorstep deliveries are restricted to a **5 km radius** around each of the 14 outlets, covering the greater Ahmedabad municipal boundaries.

9. Operational Acceptance & Sign-Off

The following commercial, operational, and clinical executives are designated to review, endorse, and accept this Business Requirements Document. Formal operational baseline endorsement remains pending final convening and sign-off.

Executive Role	Stakeholder Name	Signature & Endorsement Date	Approval Status
Principal Business Analyst & Architect	Ritesh Jadhav	Pending	Pending
Managing Director & Chain Owner	Executive Sponsor	Pending	Pending
Chief Pharmacist & Compliance VP	Dr. A. K. Patel	Pending	Pending
VP of Retail Store Operations	Retail Operations GM	Pending	Pending
VP of Commercial Strategy & Marketing	Commercial Lead	Pending	Pending

End of Document 07: Business Requirements Document (BRD)